

Task 1: RTL Design & Functional Simulation

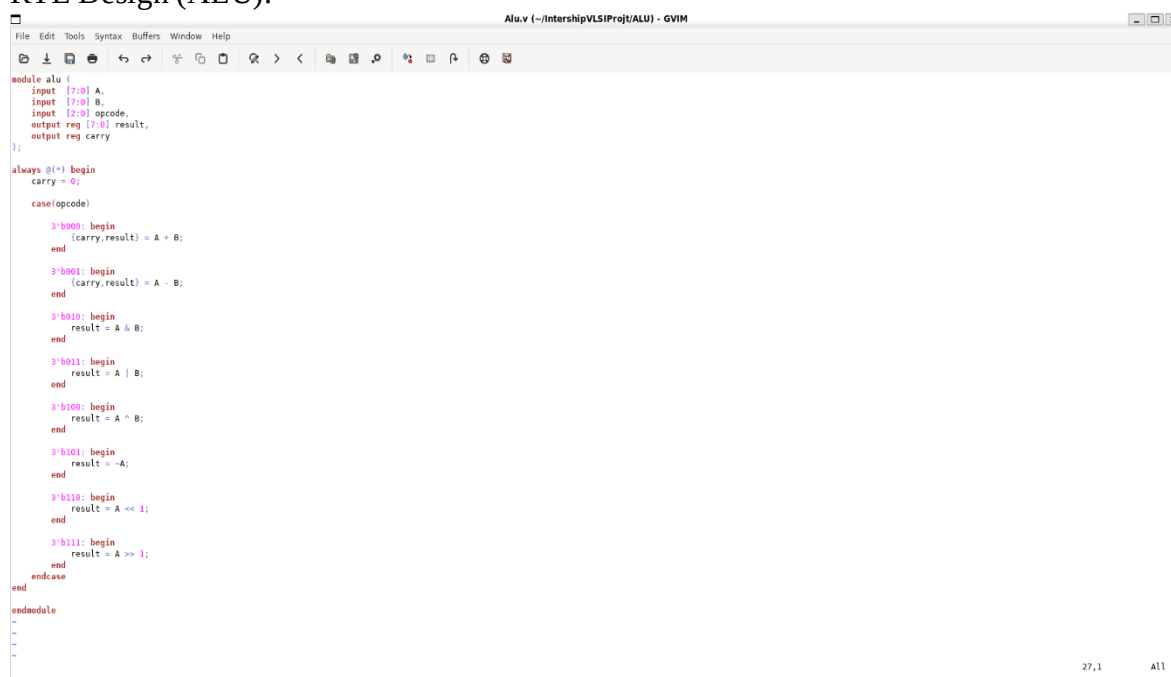
Title

8-bit ALU Design and Functional Verification using Verilog

Objective

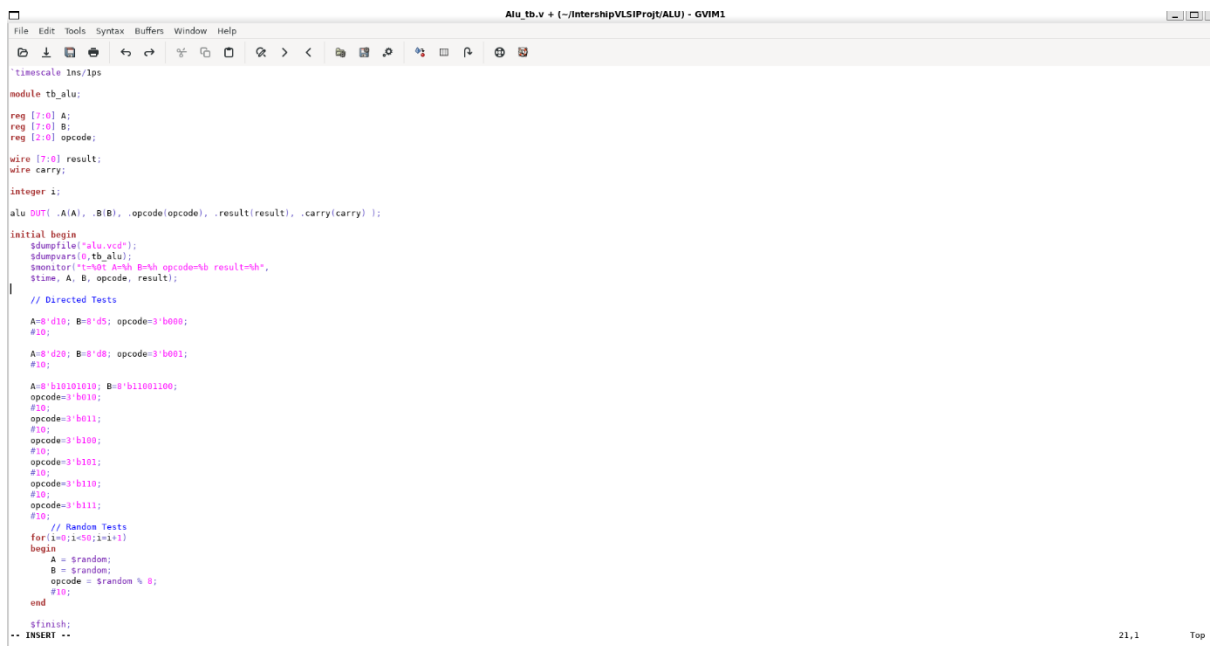
Design and verify an 8-bit ALU using Verilog HDL. Perform functional simulation using a testbench containing directed and random test cases.

RTL Design (ALU):



```
module alu (  
    input [7:0] A,  
    input [7:0] B,  
    input [2:0] opcode,  
    output reg [7:0] result,  
    output reg carry  
);  
  
always @(*) begin  
    carry = 0;  
    case(opcode)  
        3'b000: begin  
            {carry,result} = A + B;  
        end  
        3'b001: begin  
            {carry,result} = A - B;  
        end  
        3'b010: begin  
            result = A & B;  
        end  
        3'b011: begin  
            result = A | B;  
        end  
        3'b100: begin  
            result = A ^ B;  
        end  
        3'b101: begin  
            result = ~A;  
        end  
        3'b110: begin  
            result = A << 1;  
        end  
        3'b111: begin  
            result = A >> 1;  
        end  
    endcase  
end  
endmodule  
~  
~  
~
```

Testbench :

A screenshot of a text editor window titled "Alu_tb.v + (~/InterShipVLSIProjt/ALU) - GVIM1". The editor shows a Verilog testbench for an ALU. It includes module declarations, register declarations for A, B, and opcode, and wire declarations for result and carry. The testbench includes an initial block with a dumpfile, a monitor, and a series of directed tests for various operations (AND, OR, XOR, NOT, etc.) and random tests. The code is color-coded with syntax highlighting. The bottom right corner shows "21,1" and "Top".

```
Alu_tb.v + (~/InterShipVLSIProjt/ALU) - GVIM1
File Edit Tools Syntax Buffers Window Help
timescale 1ns/1ps

module tb_alu;
reg [7:0] A;
reg [7:0] B;
reg [2:0] opcode;

wire [7:0] result;
wire carry;

integer i;

alu OUT( .A(A), .B(B), .opcode(opcode), .result(result), .carry(carry) );

initial begin
    $dumpfile("alu.vcd");
    $dumpvars(0,tb_alu);
    $monitor("t=%0T A=%h B=%h opcode=%b result=%h",
            $time, A, B, opcode, result);
    // Directed Tests
    A=8'd10; B=8'd5; opcode=3'b000;
    #10;
    A=8'd20; B=8'd8; opcode=3'b001;
    #10;
    A=8'b10101010; B=8'b1001100;
    opcode=3'b010;
    #10;
    opcode=3'b011;
    #10;
    opcode=3'b100;
    #10;
    opcode=3'b101;
    #10;
    opcode=3'b110;
    #10;
    opcode=3'b111;
    #10;
    // Random Tests
    for(i=0;i<50;i=i+1)
    begin
        A = $random;
        B = $random;
        opcode = $random % 8;
        #10;
    end
    $finish;
end
-- INSERT --
21,1 Top
```

Simulation Commands:

- 1) iverilog Alu.v Alu_tb.v -o alu.vvp
- 2) vvp alu.vvp
- 3) gtkwave alu.vcd

